

# Machine Learning

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# Assessment

- Internal:External->50:50
- InternalAssessment
  - Mid-Term examination:25Marks
  - ContinuousAssessment:25Marks
    - Assignment
    - Test
    - Code-hr marks
- ExternalAssessment
  - End semester examination:100marks

# #1 Introduction

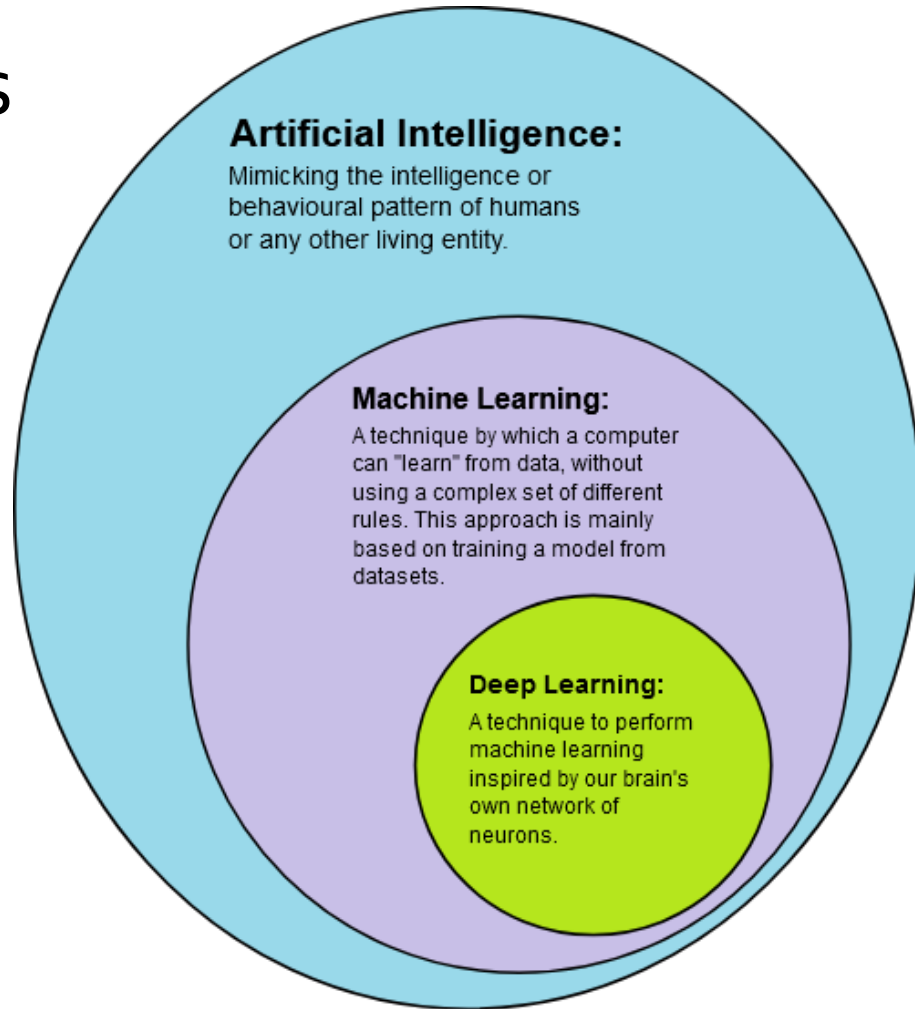
# The world we live in

- [Tesla](#)
- [Sora | OpenAI](#)
- [Godfather Extended: Deepfake Edition | Mohanlal, Mammooty, DQ and Fahadh Faasil](#)
- <https://www.facebook.com/reel/3676485885902222>
- <https://twitter.com/i/status/1851258134049837169>
- <https://www.youtube.com/watch?v=IQWUKWM2JrQ>
- chatgpt, gemini
- [Jules - An Autonomous Coding Agent](#)

- [Incredible Drone and Fireworks Show in China Welcomes New Year 2026 – World Record Event](#)
- [China's Secret Robot Army Exposed The Future of War is Here](#)

# Some Terminologies

- Artificial intelligence?
- Machine Learning?
- Deep Learning?



# What is artificial intelligence?

Artificial intelligence is the ability of a computer to perform tasks commonly associated with intelligent beings.

- See and Understand (Computer Vision)
- Speech recognition
- Decide
- Respond

# What is machine learning?

Machine learning is the study of algorithms that learn from examples and experience (data and training)

Machine Learning is the study of algorithms that

- improve their performance **P**
- at some task **T**
- with experience **E**.

A well-defined learning task is given by  $\langle P, T, E \rangle$



**This is a shirt we used to wear.**



Color: green  
Size : Large  
Type : Formal

---

**Is this a shirt ?**



Ok



Color: Green, red  
Size : Large, small  
Type : Formal,  
casual

**This is also a shirt**

# Is this a shirt ?

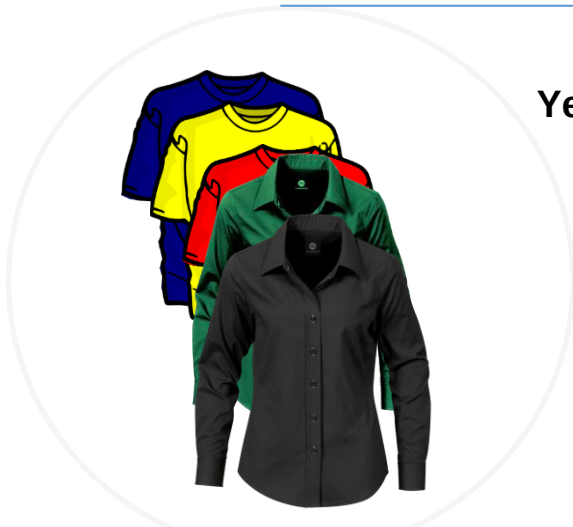


Ok



Color: Green, red, yellow  
Size : Large, small, medium  
Type : Formal, casual

## This is also a shirt



Yes, these are all shirts



I learned to  
identify every  
shirt

**What is P,T, E?**

# Machine Learning Vs traditional rule-based programming

- Traditional programming relies on human-written rules and data to output answers.
- Machine learning reverses this by taking data and answers (or outcomes) to discover the underlying rules or patterns automatically.

Net Pay = Gross Base Salary + Allowances-  
Deductions

$$\text{Net Pay} = (B + (w_H \cdot B) + (w_D \cdot B) + (w_T \cdot B)) - D$$

# Applications of Machine Learning

- Machine learning addresses problems where **explicit programming is difficult**:
  - Pattern recognition
  - Prediction and forecasting
  - Classification and clustering
  - Decision making under uncertainty
  - Anomaly detection
- Examples: spam detection, disease diagnosis, image recognition.

# Data

- Data is the **core component** of ML systems.
- **We train the models using data**
- **Types of data:**
  - Structured (tables, databases)
  - Unstructured (images, text, audio)
  - Semi-structured (logs, JSON)
- **Label types:**
  - Labeled (supervised learning)
  - Unlabeled (unsupervised learning)
  - Partially labeled (semi-supervised)
- Data partitioning
  - Training set
  - Validation set (optional)
  - Test set

Quality, quantity, and diversity of data strongly influence performance.

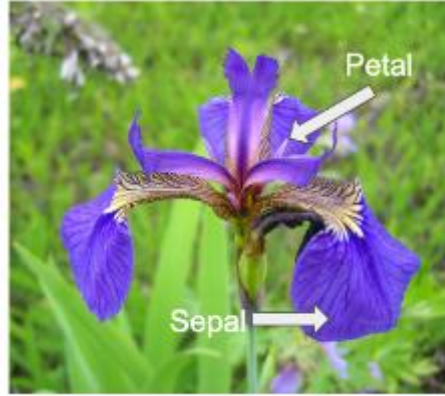
Dataset: Student Pass Prediction

|          |
|----------|
| Features |
| Label    |

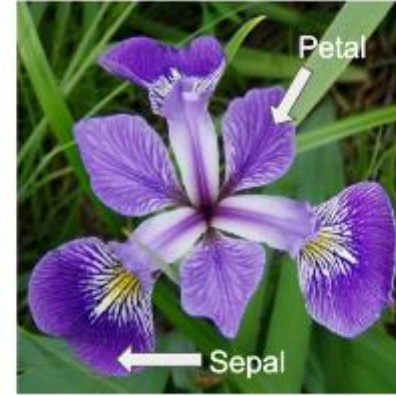
| Student_ID | Study_Hours | Attendance (%) | Assignments_Completed | Result |
|------------|-------------|----------------|-----------------------|--------|
| 1          | 2           | 60             | 3                     | Fail   |
| 2          | 3           | 65             | 4                     | Fail   |
| 3          | 4           | 70             | 5                     | Pass   |
| 4          | 5           | 75             | 6                     | Pass   |
| 5          | 1           | 50             | 2                     | Fail   |
| 6          | 6           | 90             | 8                     | Pass   |
| 7          | 7           | 95             | 9                     | Pass   |
| 8          | 2           | 55             | 3                     | Fail   |

# Example

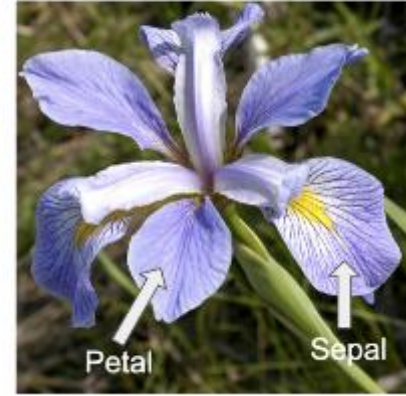
*Iris setosa*



*Iris versicolor*



*Iris virginica*



## Labelled Dataset

## Iris Flower species

|    | Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species    |
|----|--------------|-------------|--------------|-------------|------------|
| 1  | 5.1          | 3.5         | 1.4          | 0.2         | setosa     |
| 2  | 4.9          | 3.0         | 1.4          | 0.2         | versicolor |
| 3  | 4.7          | 3.2         | 1.3          | 0.2         | setosa     |
| 4  | 4.6          | 3.1         | 1.5          | 0.2         | setosa     |
| 5  | 5.0          | 3.6         | 1.4          | 0.2         | versicolor |
| 6  | 5.4          | 3.9         | 1.7          | 0.4         | setosa     |
| 7  | 4.6          | 3.4         | 1.4          | 0.3         | Virginica  |
| 8  | 5.0          | 3.4         | 1.5          | 0.2         | setosa     |
| 9  | 4.4          | 2.9         | 1.4          | 0.2         | versicolor |
| 10 | 4.9          | 3.1         | 1.5          | 0.1         | versicolor |



# Types of Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Semi supervised Learning

# Types of Machine Learning

## a) Supervised Learning

- In Supervised learning a model is trained on a labeled dataset (training set).
- This means that for every training example, the algorithm is provided with both the input data (features) and the correct corresponding output (the ground truth label).
- The primary goal of supervised learning is to find a mapping function **f** that can accurately predict the output **y** for new, unseen input data **x**:

$$y = f(x)$$

- Supervised learning methods
  - Regression
  - Classification
    - K Nearest Neighbour (KNN)
    - Support Vector Machines (SVM)
    - Decision Tree
    - Random Forest
    - **Artificial Neural Networks**

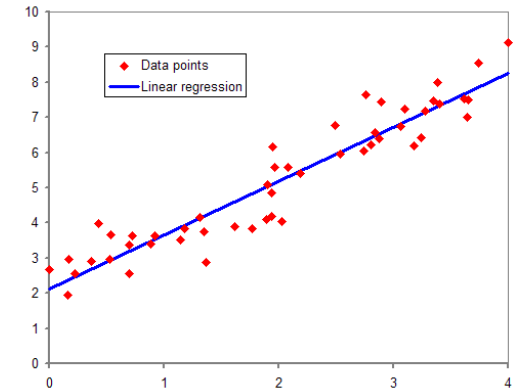
- Regression (predicts continuous values)

### Example: House Price Prediction

Target (Y): House Price

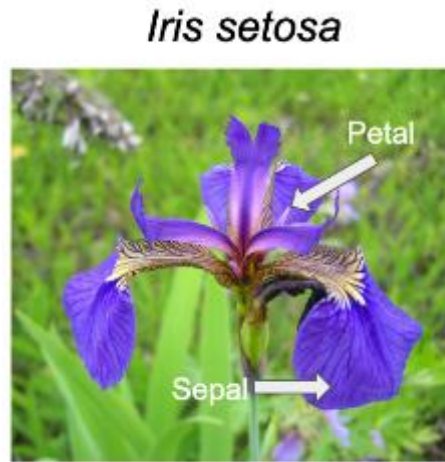
Input features (X): House Size, Number of Bedrooms

| House_Size_sqft | Bedrooms | House_Price (₹ Lakhs) |
|-----------------|----------|-----------------------|
| 600             | 1        | 25                    |
| 800             | 2        | 35                    |
| 1000            | 2        | 45                    |
| 1200            | 3        | 60                    |
| 1500            | 3        | 75                    |
| 1800            | 4        | 95                    |
| 2000            | 4        | 110                   |
| 2300            | 5        | 130                   |



# Classification:

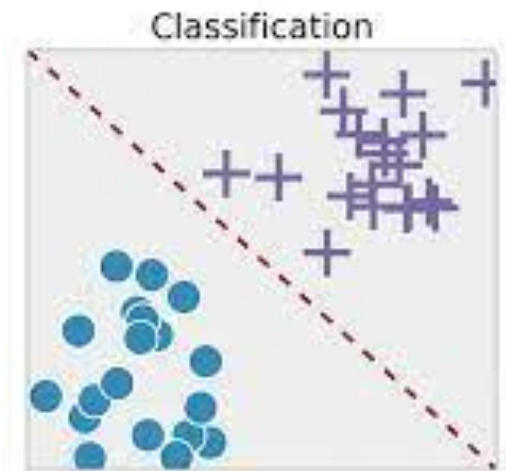
(Predicts categorical labels)



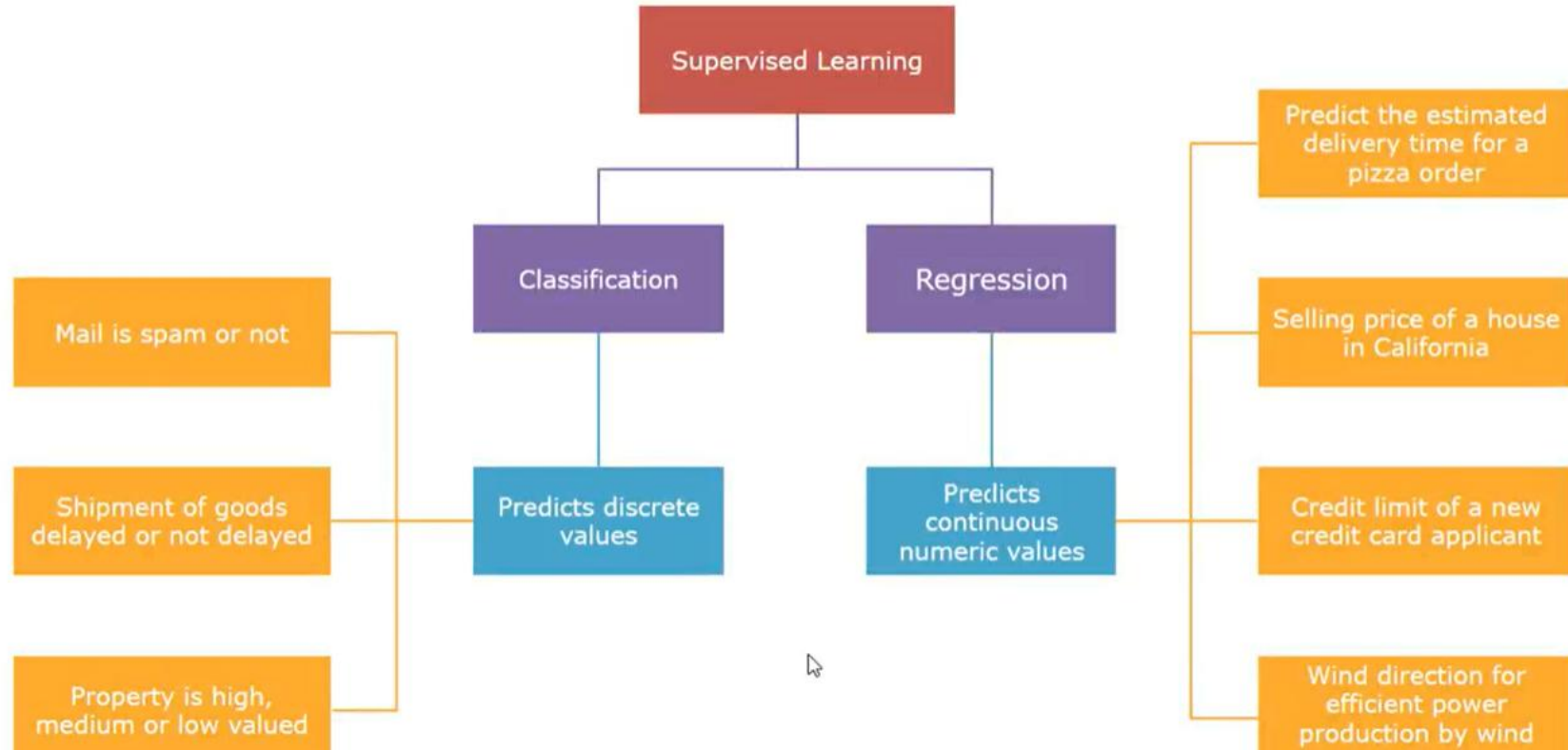
## Labelled Dataset

## Iris Flower species

|    | Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species    |
|----|--------------|-------------|--------------|-------------|------------|
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| 5  | 5.0          | 3.6         | 1.4          | 0.2         | versicolor |
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| 7  | 4.6          | 3.4         | 1.4          | 0.3         | Virginica  |
| 8  | 5.0          | 3.4         | 1.5          | 0.2         | setosa     |
| 9  | 4.4          | 2.9         | 1.4          | 0.2         | versicolor |
| 10 | 4.9          | 3.1         | 1.5          | 0.1         | versicolor |

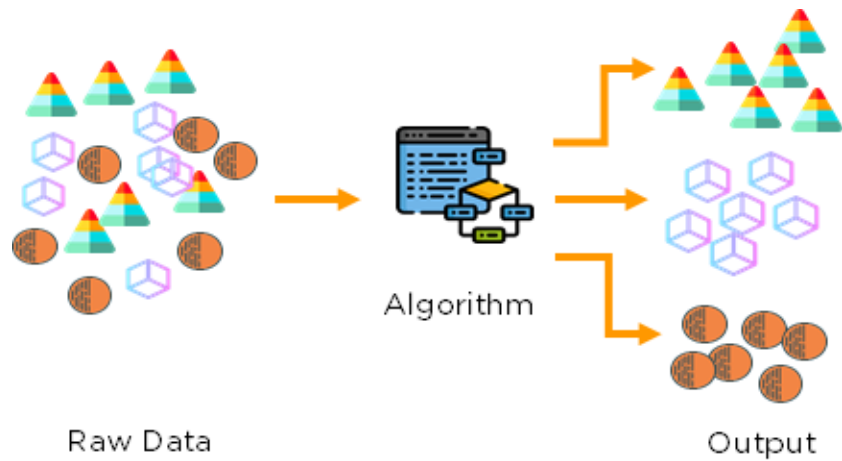


# Types of Supervised Learning



## b) Unsupervised Learning

- Unsupervised learning is a type of machine learning where an algorithm is trained on data that has **not been labeled, categorized, or annotated**.
- It analyzes the dataset to discover hidden patterns, underlying structures, or natural groupings entirely on its own.
- **Unsupervised Learning methods**
  - Clustering
  - Association rule mining
  - Dimensionality reduction



Clustering

| ID  | Items                                        |
|-----|----------------------------------------------|
| 1   | {Bread, Milk}                                |
| 2   | {Bread, <b>Diapers</b> , <b>Beer</b> , Eggs} |
| 3   | {Milk, <b>Diapers</b> , <b>Beer</b> , Cola}  |
| 4   | {Bread, Milk, <b>Diapers</b> , <b>Beer</b> } |
| 5   | {Bread, Milk, Diapers, Cola}                 |
| ... | ...                                          |

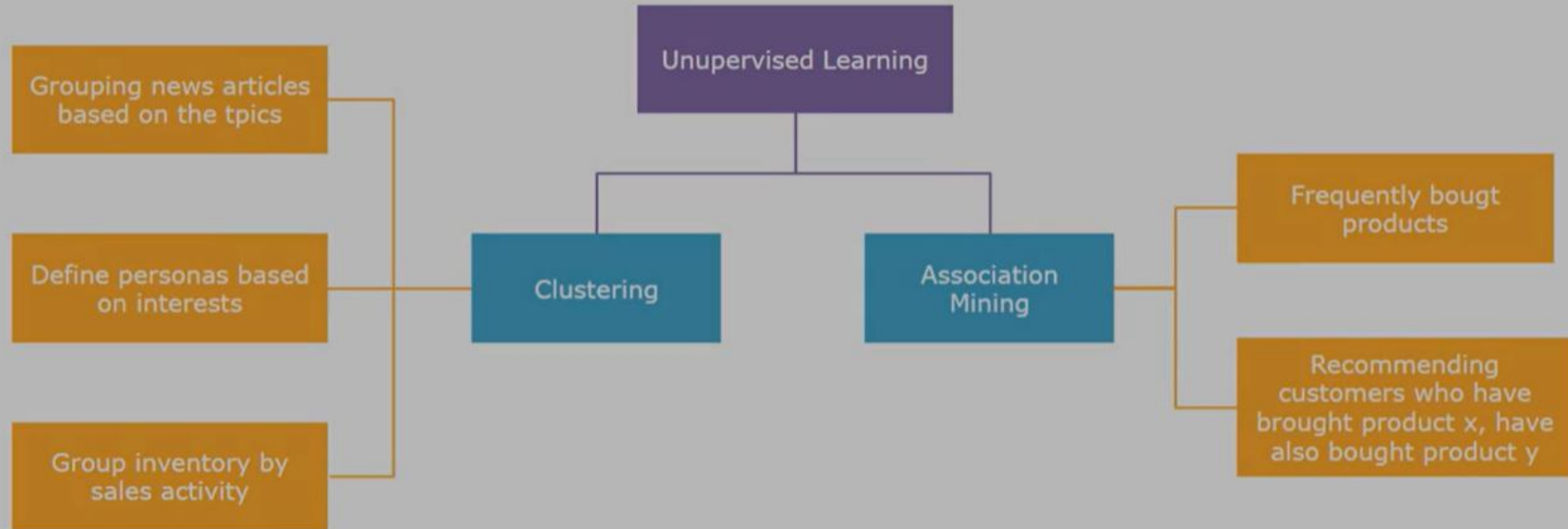
market  
basket  
transactions

**{Diapers, Beer}** Example of a frequent itemset

**{Diapers} → {Beer}** Example of an association rule

Association rule mining

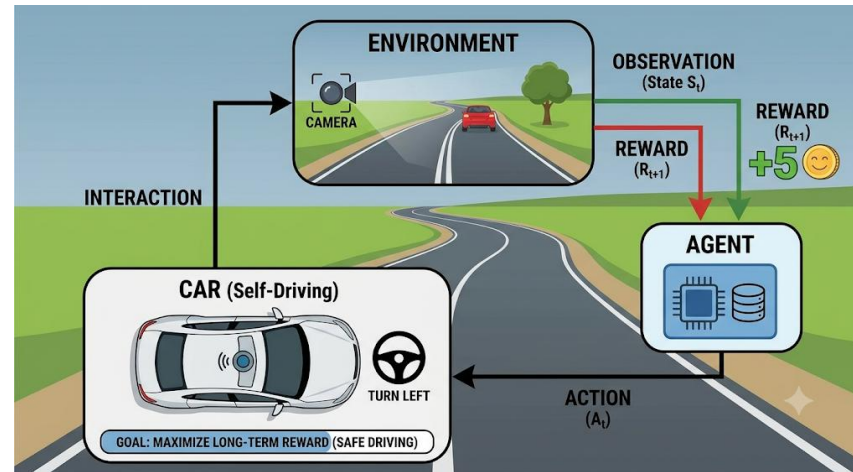
# Types of Unsupervised Learning





## c) Reinforcement learning

- Reinforcement learning (RL) is a branch of machine learning where an **autonomous agent** learns to make decisions by interacting with an environment.
- Used when training data is unavailable
- The agent interacts with an environment through trial and error. It takes actions, transitions to new states, and receives feedback in the form of **rewards or penalties**, aiming to maximize its cumulative reward over time by better learning
- The agent **creates its own dataset** while learning (state, action, reward, next state).
- Example- self-driving car

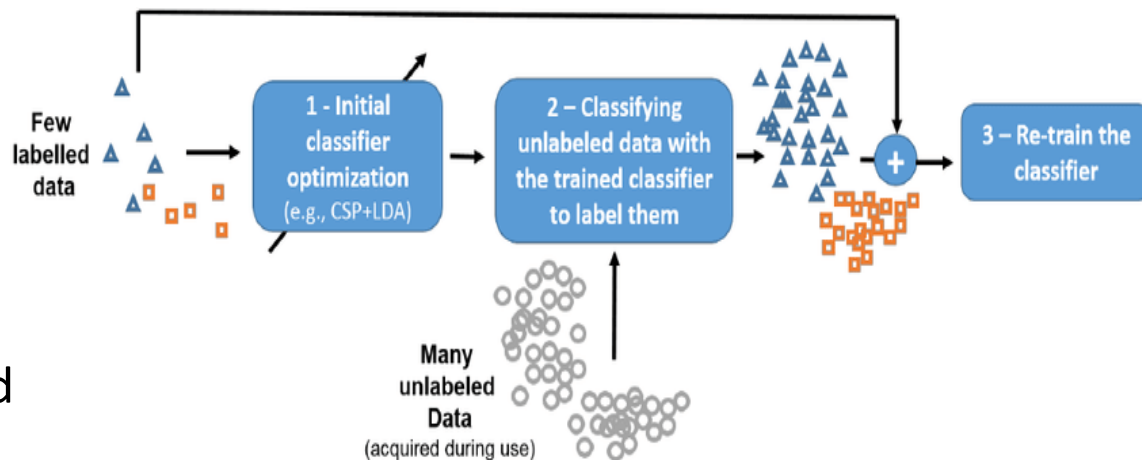


## D) SEMI SUPERVISED LEARNING

Semi-supervised learning is a machine learning approach that uses both labeled and unlabeled data for training. It is especially useful when labeled data is scarce or expensive, but unlabeled data is abundant.

### How Semi-Supervised Learning Works

1. Train a model using the labeled data
2. Predict labels for unlabeled data
3. Select confident predictions
4. Retrain the model using both real and predicted labels



# Core aspects of developing a learning system

- Problem Formulation and Data Collection
- Data Preprocessing and Feature Engineering
- Data Splitting (The Partition Strategy)
  - Training set, validation Set (Optional), Test set
- Model Selection and Training
- Evaluation and Diagnostics

# Supervised Learning Model (Classification)

## Input features:

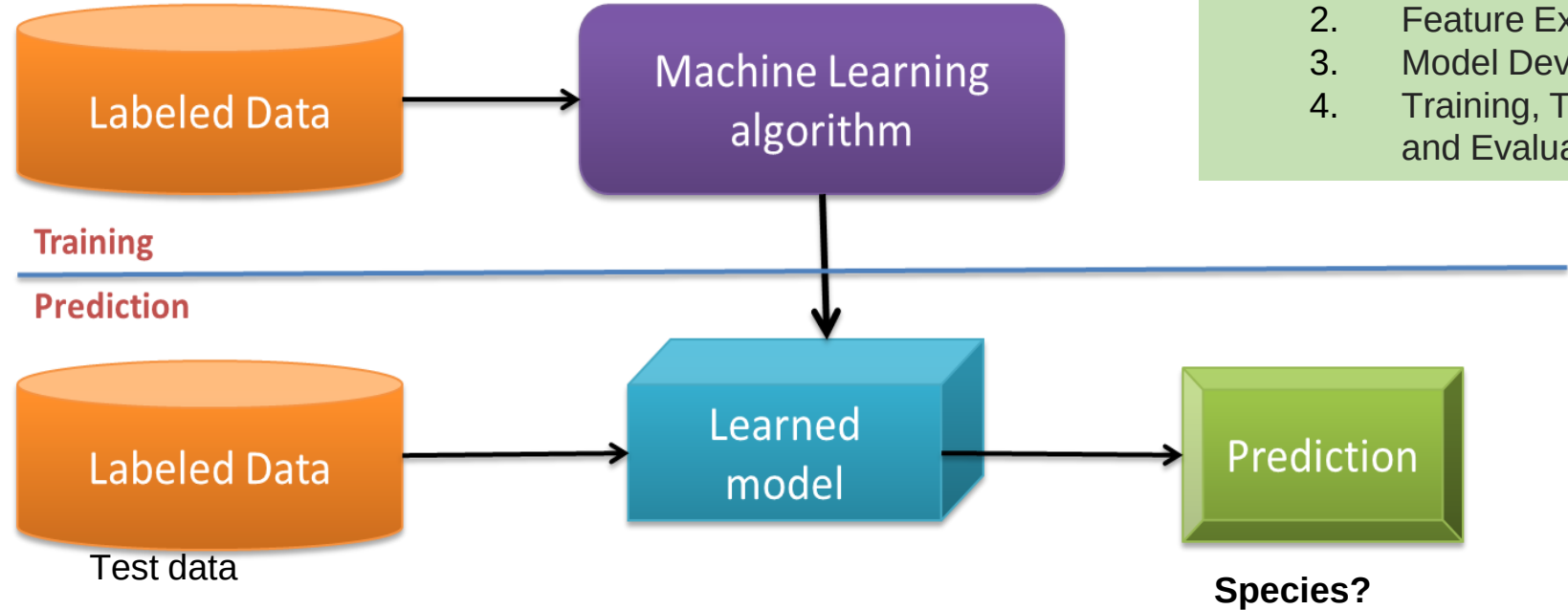
Sepal length, Sepal width, Petal length, petal width

## Label:

Iris species

## General methodology

1. Data Collection and Preprocessing
2. Feature Extraction
3. Model Development
4. Training, Testing and Evaluation



| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width |
|--------------|-------------|--------------|-------------|
| 5.1          | 3.5         | 1.4          | 0.2         |
| 4.9          | 3.0         | 1.4          | 0.2         |
| 4.7          | 3.2         | 1.3          | 0.2         |
| 4.6          | 3.1         | 1.5          | 0.2         |
| 5.0          | 3.6         | 1.4          | 0.2         |

# Supervised Learning Model (Classification)

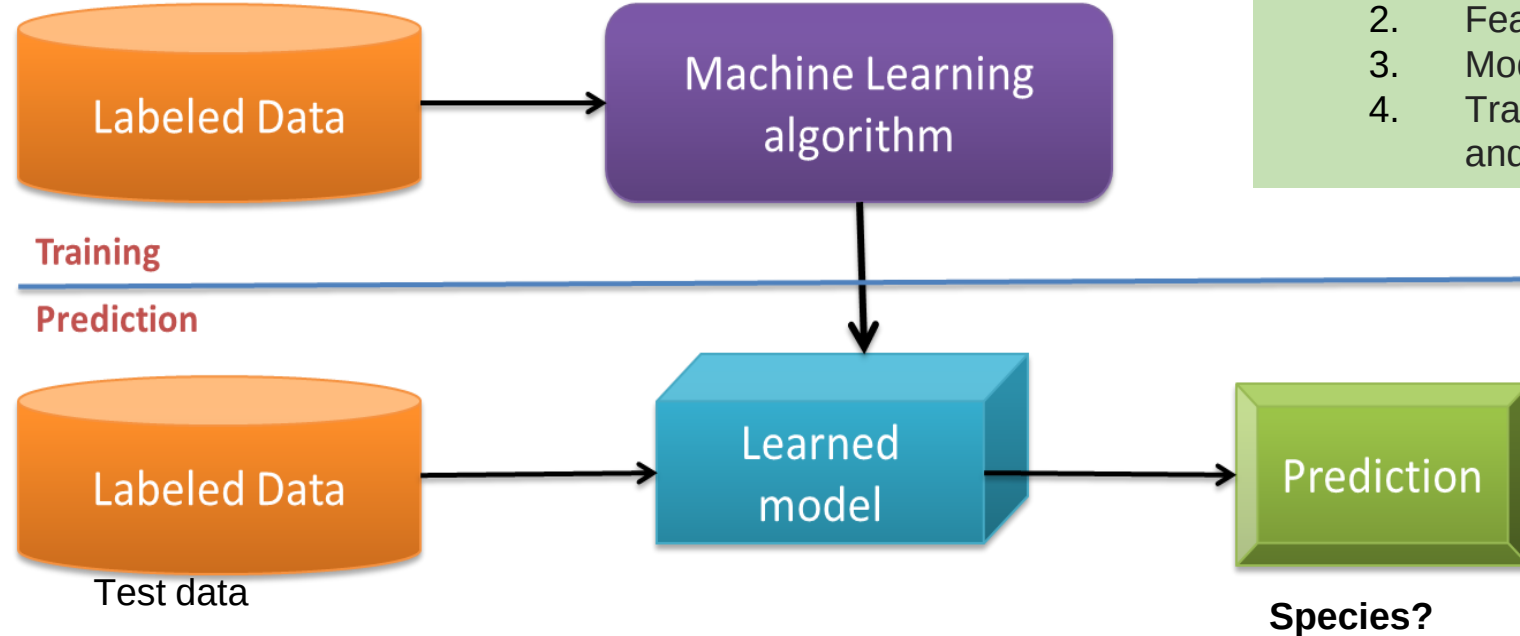


## Input features:

Sepal length, Sepal width, Petal length, petal width

## Label:

Iris species



## General methodology

1. Data Collection and Preprocessing
2. Feature Extraction
3. Model Development
4. Training, Testing and Evaluation

Is the prediction correct?



# Supervised Learning Model (Classification)

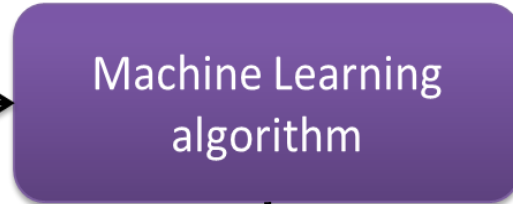


## Input features:

Sepal length, Sepal width, Petal length, petal width

## Label:

Iris species

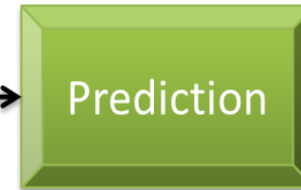
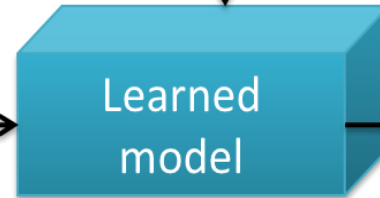


Training

Prediction



Test data



Species?

## General methodology

1. Data Collection and Preprocessing
2. Feature Extraction
3. Model Development
4. Training, Testing and Evaluation

## Evaluation metrics

Accuracy  
Precision  
Recall  
.....



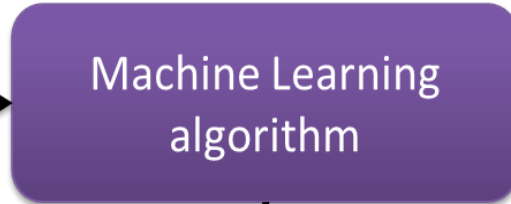
# Supervised Learning Model (Classification)

## Input features:

Sepal length, Sepal width, Petal length, petal width

## Label:

Iris species

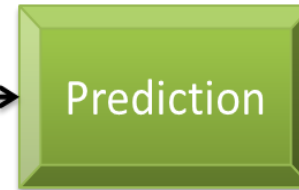
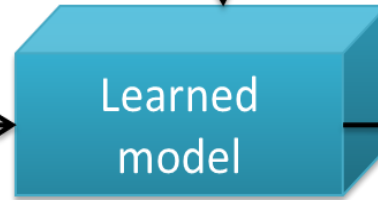


Training

Prediction



Test data



Species?

**Evaluation metrics**  
Accuracy=98%

## General methodology

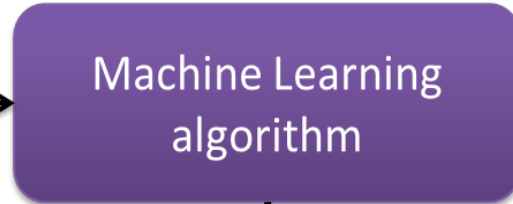
1. Data Collection and Preprocessing
2. Feature Extraction
3. Model Development
4. Training, Testing and Evaluation

98% of the predictions by the model are correct!!



# Supervised Learning (Classification)

↓  
**Input features:**  
Sepal length, Sepal width, Petal length, petal width  
**Label:**  
Iris species



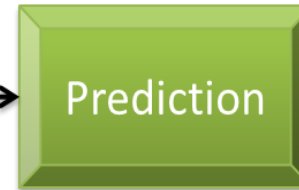
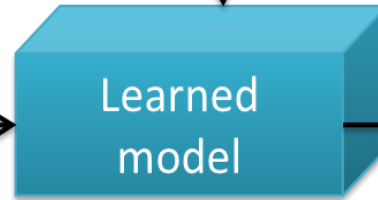
- General methodology**
1. Data Collection and Preprocessing
  2. Feature Extraction
  3. Model Development
  4. Training, Testing and Evaluation

**Training**

**Prediction**



Test data

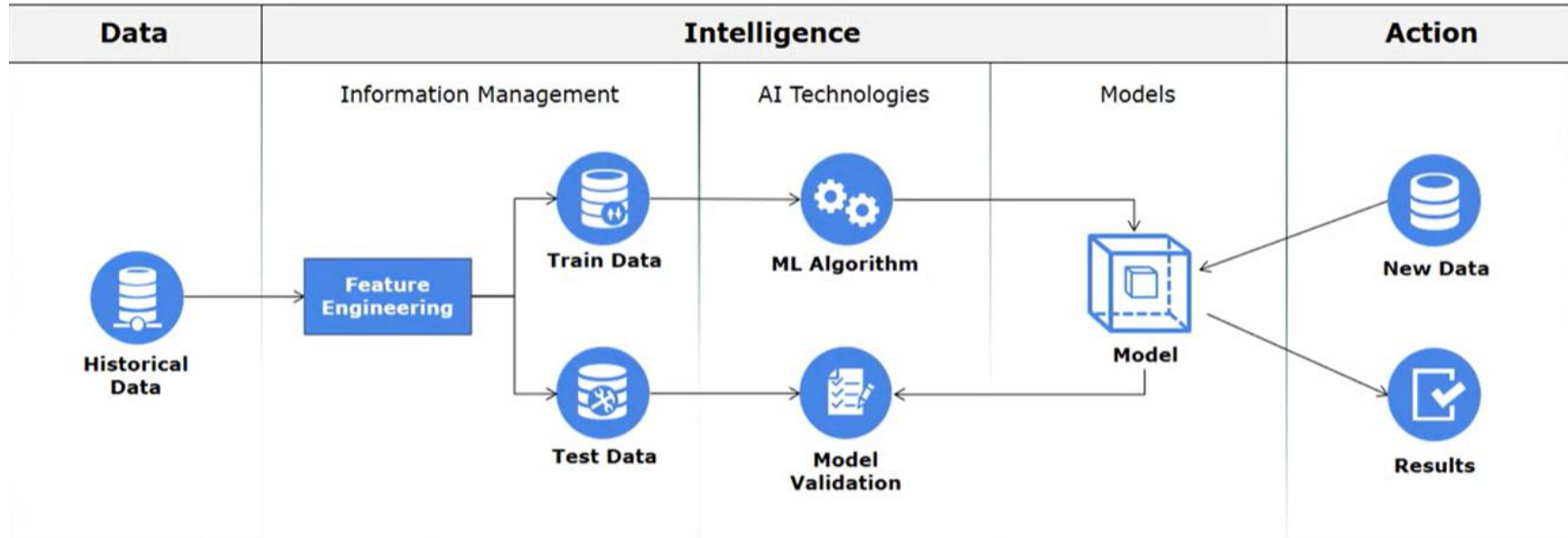


**Species?**

↑  
**New data**  
**Input features:**  
[6, 3.2, 3.6, 5.1]



## Putting the concepts together: ML process



Historical data from various sources are cleaned and subjected to feature Engineering where significant features are selected. This engineered data is divided into two sets: Train data and Test data. Machine learning models are built using train data which are validated using test data. This validated model is used for prediction.

# When to use machine learning?

ML is used when:

- a. Human expertise does not exist (navigating on Mars)
- b. Humans can't explain their expertise (speech recognition)
- c. Models must be customized (personalized medicine)
- d. Models are based on huge amounts of data (genomics)

Learning isn't always needed - There is no need to "learn" to calculate payroll